Foundation of Artificial Intelligence

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field that attempts not only to understand how humans think, perceive, and act, but also to build intelligent entities capable of performing those functions. The authors characterise AI as a "universal field" because it is relevant to any intellectual task; it encompasses a vast range of activities that require cognitive effort, spanning the spectrum from general mechanisms to highly specific applications.

The breadth of the discipline is demonstrated through its diverse subfields, which range from:

- **General Tasks:** Such as the study of learning and perception.
- **Specific Challenges:** Including playing chess, proving mathematical theorems, and writing poetry.
- **Physical and Professional Applications:** Such as driving a car on a crowded street and diagnosing diseases.

Important Characteristics of AI

AI systems can perform several activities associated with human intelligence:

1. **Learning** – learning patterns from data.
2. **Reasoning** – using information to reach conclusions.
3. **Problem-solving** – finding solutions to problems.
4. **Decision-making** – selecting an appropriate action based on available information.
5. **Perception** – understanding images, sounds, and other inputs.
6. **Language understanding** – processing and generating human language.
7. **Prediction** – estimating what may happen in the future.

The Four Approaches to AI: A Categorisation

The definitions of AI can be organised into four distinct approaches based on whether they focus on thought processes versus behaviour, and whether success is measured by fidelity to human performance versus an ideal standard of rationality.

Process / Goal	Human-centred	Rationalist (Ideal)
Thought	**Thinking Humanly:** Focuses on cognitive modelling and the internal mechanisms of the mind.	**Thinking Rationally:** Focuses on the "laws of thought" and logical inference.
Behaviour	**Acting Humanly:** Focuses on the ability to perform functions that require intelligence if done by people.	**Acting Rationally:** Focuses on the design of intelligent agents that achieve the best expected outcome.

Acting Humanly: The Turing Test Approach

This approach focuses on the operational definition of intelligence provided by the Turing Test, where a computer must be indistinguishable from a human in its responses. While the standard test requires natural language processing, knowledge representation, automated reasoning, and machine learning, the **total Turing Test** adds computer vision and robotics to test perceptual and physical abilities. Success is measured by the system's ability to act like an exemplar human, providing the justification for including sensory and motor capabilities in AI research.

Thinking Humanly: The Cognitive Modelling Approach

To build a system that thinks like a human, one must first determine how humans think through introspection, psychological experiments, or brain imaging. Once a theory of the mind is established, it can be expressed as a computer programme; if the programme's input–output behaviour matches human behaviour, it provides evidence that the programme's mechanisms may also be operating in the human mind. This interdisciplinary effort constitutes cognitive science.

Thinking Rationally: The "Laws of Thought" Approach

Drawing on the Aristotelian tradition, this approach focuses on codifying "right thinking" through irrefutable reasoning processes known as **syllogisms**. The goal is to develop a formal notation for statements about the world and their relations, allowing a system to derive new conclusions logically from existing premises. Success is measured by the system's ability to solve any solvable problem described in logical notation, though this approach faces challenges when dealing with informal or uncertain knowledge.

Acting Rationally: The Rational Agent Approach

This approach defines AI as the study of the design of intelligent agents that act so as to achieve the best outcome (or the best expected outcome under uncertainty). Success is measured against a mathematically well-defined standard of rationality. This is more general than the "laws of thought" approach because correct inference is only one possible mechanism for achieving rational behaviour, and it is more amenable to scientific development than approaches based on the sum total of human behaviour.

The Foundations of AI (Interdisciplinary Roots)

Modern AI is built upon the contributions of eight foundational disciplines, each addressing specific questions regarding the nature of intelligence and computation.

1. **Philosophy:** Asks "How does knowledge lead to action?" It contributed the development of logic, the distinction between mind and matter (dualism vs materialism), and the theory that actions are justified by the logical connection between goals and knowledge.
2. **Mathematics:** Asks "What can be computed?" It provided the formal rules for logic, the definition of algorithms, the concept of tractability (NP-completeness), and the theory of probability for reasoning with uncertainty.

3. **Economics:** Asks "How should we make decisions when the payoff may be far in the future?" It contributed decision theory, game theory for multiagent environments, and the concept of "satisficing"—making decisions that are good enough rather than optimal.
4. **Neuroscience:** Asks "How do brains process information?" It established that the brain is the seat of consciousness, composed of neurons that process signals; notably, while computer cycle times are roughly 10^{-9} seconds, neuron cycle times are much slower at 10^{-3} seconds, explaining why the brain relies on massive parallelism.
5. **Psychology:** Asks "How do humans and animals think and act?" It established the cognitive view of the brain as an information-processing device and promoted the idea that a cognitive theory should be expressed as a detailed computer programme.
6. **Computer Engineering:** Asks "How can we build an efficient computer?" It provided the physical artifact (the computer) required for AI to succeed, delivering the necessary speed and storage tools for intelligence to be manifested.
7. **Control Theory and Cybernetics:** Asks "How can artifacts operate under their own control?" It contributed the idea of purposive behaviour arising from regulatory mechanisms and feedback loops to achieve stable, adaptive behaviour by minimising "error."
8. **Linguistics:** Asks "How does language relate to thought?" It provided the basis for modern natural language processing by showing that language learning involves creativity and formal structures that can be programmed, leading to the field of computational linguistics.

The Rational Agent Paradigm

The rational agent paradigm provides a unifying framework for the study of AI. An agent is defined as the coupling of an **architecture** (the physical sensors and actuators) and a **programme** (the mapping from percepts to actions).

Understanding Rationality

Rationality is distinguished from **omniscience** and **perfection**. While an omniscient agent knows the actual outcome of its actions, rationality only requires an agent to maximise *expected* performance based on the evidence provided by its percept sequence and built-in knowledge. Rationality depends on four factors:

- The performance measure defining success.
- The agent's prior knowledge of the environment.
- The actions the agent can perform.
- The agent's percept sequence to date.

Autonomy and Learning

Autonomy is critical for a rational agent; an agent lacks autonomy if it relies solely on the prior knowledge of its designer. To be truly autonomous, an agent must be able to learn from its environment to compensate for partial or incorrect prior knowledge. Over time, the incorporation of learning allows an agent to become effectively independent of its initial configuration through experience.

Task Domains and AI Capability

AI tasks and systems are categorised into three primary domains based on the nature of the challenges they address and the specialised knowledge required.

Formal Tasks

These involve structured problems with clear rules and logical notation.

- **Chess:** The **DEEP BLUE** system became the first to defeat a world champion in a match.
- **Mathematical Theorems:** Systems like the Geometry Theorem Prover were among the earliest successes.

Mundane Tasks

These involve everyday abilities such as perception and navigation that are complex for machines to replicate.

- **Computer Vision and Robotics:** The autonomous car **STANLEY** won the DARPA Grand Challenge, while **BOSS** won the Urban Challenge by navigating traffic and obeying rules.
- **Autonomous Planning:** NASA's **REMOTE AGENT** was the first on-board autonomous planning programme to control a spacecraft, while **MAPGEN** and **MEXAR2** have handled daily mission planning for Mars exploration.

Expert Tasks

These require specialised professional knowledge and have been the focus of commercial AI.

- **Medical Diagnosis:** Systems like **MYCIN** demonstrated expert-level performance in diagnosing blood infections.
- **Logistics Planning:** The **DART** tool was used for automated transportation scheduling during the Persian Gulf crisis.
- **Expert Systems:** The **R1** system helped configure computer orders, saving millions of pounds annually.

The PEAS Framework for Task Environment Specification

The PEAS framework is used to fully specify the task environment of an agent. It stands for Performance, Environment, Actuators, and Sensors.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi Driver	Safety, speed, legality, comfort, profit	Roads, traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, GPS, speedometer, engine sensors

Medical Diagnosis	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Part-picking Robot	Percentage of parts in correct bins	Conveyor belt, parts, bins	Jointed arm and hand	Camera, joint angle sensors

Applications of Artificial Intelligence (AI)

Artificial Intelligence is a rapidly growing technology that is penetrating the contemporary world through solutions to numerous problems within different sectors. Taking health care, education, finance, entertainment, and agriculture as some of the sample sectors, there is significant improvement in efficiency, accuracy, and convenience from the general application of artificial intelligence. It replicates human thinking abilities and can solve problems concerning learning, reasoning, and decision-making much more proficiently than a human being.

From helping cars to drive themselves, enhancing the experience a user gets on an interface, or helping researchers in their quest for new knowledge about our universe, AI is asserting itself in the age of smart systems. Thus, it is significant to learn more and use this impactful technology correctly.

AI Applications

The following are some sectors that apply artificial intelligence.



AI in Astronomy

- **Automated Celestial Object Identification:** Stars, Galaxies, and space phenomena Identification from telescope image data with a high level of accuracy. This speaks to the discovery of new phenomena and datasets that reduce the role of astronomers in making observations on what they find fascinating.
- **Exoplanet Hunting:** It uses the variation in the brightness of stars to look for drops that suggest the presence of planets orbiting the stars, assisting scientists in finding new exoplanets.
- **Analyzing Space Information:** AI receives large data feeds from telescopes and makes patterns or irregularities that help in discoveries, such as the confirmation of dark matter or black holes.
- **Real-time Monitoring of Space Events:** AI systems are used to constantly survey space and look for such events as supernovae to alert astronomers.
- **Intelligent Telescope Control:** AI sets telescope characteristics according to the actual climate or other conditions or depending on the type of observations that are to be conducted to optimize the images and timeliness of receiving them.

AI in Healthcare

- **Medical Imaging Analysis:** Diagnosing patients' condition, interpreting X-rays, MRI or CT scans to identify cancerous tissue, breakages, internal bleeding & more.
- **Predictive Diagnostics:** Patients' history and habits are used to anticipate diseases and conditions before occurrence, and this helps in the diagnosis of diseases such as heart ailments or cancer.
- **Drug Discovery & Development:** Here, the software mimics the molecular behaviors to screen and select potent drug molecules in a shorter period and with less investment than the corresponding costs involved in the conventional approach.
- **Personalized Medicine:** AI takes into consideration the patient's genes and medical history to ensure the best results with minimal side effects.
- **Operational Efficiency in Hospitals:** AI enhances the usage of resources, patient traffic, and staff timetables, as well as limits the wait time for patients.

AI in Gaming

- **Smart NPC Behavior:** AI makes non-player characters react and adapt dynamically, providing a challenging and immersive gaming experience.
- **Procedural content generation:** Another area of AI application in games is procedural content generation since it is a time-saving invention that offers a wide variety of game environments, levels, and quests for the players.
- **Realistic Graphics and Physics:** AI improves the graphics, which makes characters and objects look almost real, as well as making the physical aspects of the game real.

AI in Finance

- **Fraud Detection:** Through machine learning, AI can analyze and quickly distinguish fraudulent transactions based on previous instances, lowering the occurrence of fraud.
- **Algorithmic Trading:** Accurate computer-driven buying and selling of stocks in the market with high rates of gains, which determines the best time to buy and sell depending on the current data and models.
- **Credit Risk Assessment:** The AI here looks at how a borrower handles their credit; this enables the lenders to make informed decisions.

AI in Data Security

- **Anomaly detection:** Anomaly detection deals with mining the traffic and identifying exceptions, which may indicate potentially problematic issues such as breaches or malware.
- **Risk Estimation:** Based on the past records of attacks, the AI determines the possible threats in the future for which preparations can be made beforehand.
- **Automated Response Systems:** When an invasion is identified, it is possible for an AI to shut down the affected computers or launch a counterattack immediately.

AI in Social Media

- **Content Recommendations:** It follows the behavior and habits of its users and makes suggestions for posts, videos, or advertisements that entice the users.
- **AI Chatbots:** These bots automate customer service and avoid delays by relying on customers' messages or comments.
- **Sentiment Analysis:** AI derives the attitude of users on topics or products and services and comments, which helps in business decisions.
- **Trend Identification:** AI provides brands with details of topics to look out for concerning social media content or which content is likely to go viral.

AI in Travel & Transport

- **Route Planning:** AI determines effective and efficient routes across junctions and highways based on real-time occurrence and avoids unnecessary time wastages and fuel costs.
- **Security Screening:** Intelligent scanning tools that assist in scanning eliminate threats at airports while cutting through the time it takes for security checks.
- **Virtual Travel Assistants:** AI takes the form of agents that can help with bookings, 0provide suggestions, or answer inquiries, thus increasing comfort.
- **Vehicle and Infrastructure Management:** By using AI to foresee when vehicles or infrastructures need to be repaired, there is less risk of breakdowns and accidents with preserved safety.

AI in Automotive Industry

- **Autonomous Vehicles:** In this case, AI allows the cars to sense the environment and control them by themselves.

- **Driver Assistance Systems (ADAS):** Some smart features include the ability to identify lanes, maintain lane position, use adaptive cruise control, and automatically brake in case there is an obstacle.
- **Manufacturing Automation:** AI checks for defects on production lines, controls inventories and increases the efficiency of the manufacturing factory.
- **Voice Control:** AI ensures that drivers do not need to use their hands to operate the car's navigation system, make phone calls, and control the media since this can distract their attention.

AI in Robotics

- **Autonomous Navigation:** Robots have the ability to move around and operate and work independently, for instance, in the warehouse or disaster-affected regions.
- **Object Manipulation:** AI allows robots to identify, pick, and appropriately interact with various objects for applications such as logistics and production.
- **Human-Robot Collaboration:** AI enhances the capability of the robot to be used with ease by people with the aim of assisting humans in the completion of tasks without affecting their safety at the workplace.

AI in Entertainment

- **Personalized Content:** Movies, series or music suggestions depending on the user's preferences increase the level of satisfaction among the audience.
- **Creative Tools:** One of the most important creative use cases of AI is that AI provides the tools to support artists as co-creators in the production of pieces such as music, artwork, or videos.
- **Interactive Live Shows:** Through AI, real-time translations on stage shows and the overall impact that is exhibited during the show.

AI in Agriculture

- **Crop Monitoring:** Drones and sensors help the AI system detect crop health, moisture, and pests to enable timely action.
- **Precision Agriculture:** AI knows how much water, fertilizer, or pesticide is best to apply to a certain area in order to have maximum return without wasting resources.
- **Automated Equipment:** In the system used for planting, applying chemicals, and even reaping, machines are driven by AI technology; thus, the cost of labor is high.
- **Livestock Tracking: The** AI system is used in animals to help farmers note the health or any abusive behavior associated with the stock.

AI in E-commerce

- **Product Recommendations:** By using product recommendations, the customer is offered products related to their interests, thus making it easy for sales to be made and the customer satisfied with the recommended products.
- **Inventory Management:** AI anticipates customer demand and can automatically reorder to replenish stocks or else order too many products that are not in high demand.

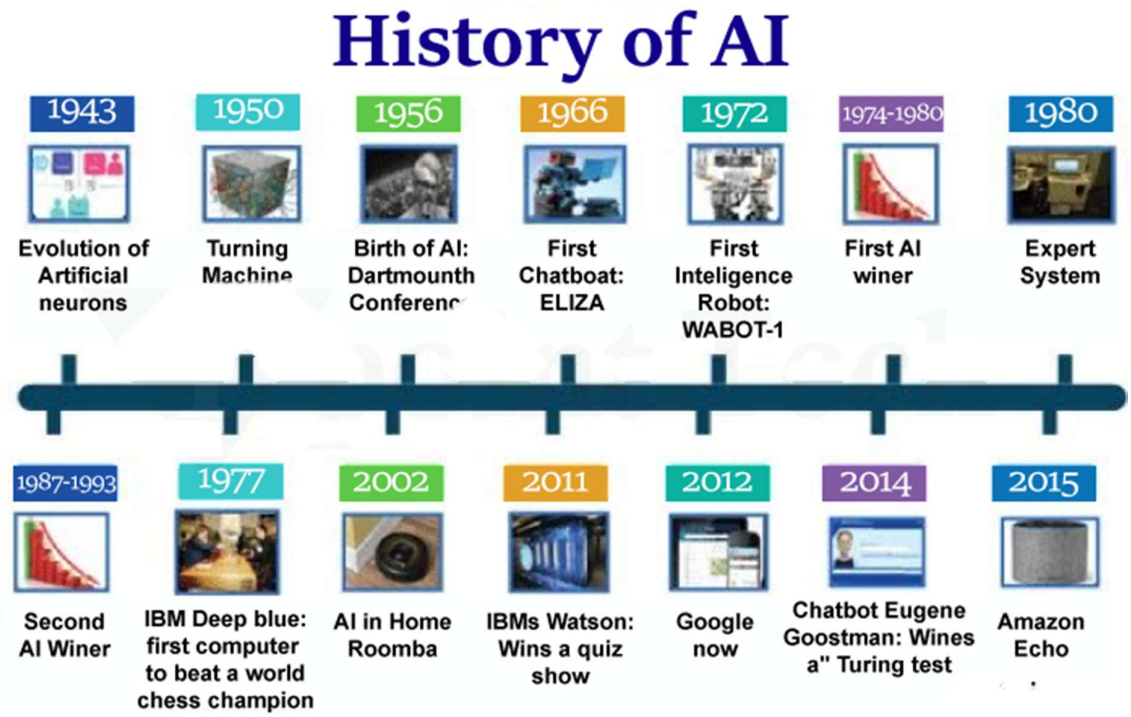
- **Dynamic Pricing:** Prices are changed periodically according to factors influencing them, such as market trends, competitor prices, and customer demand, to extract the maximum amount of money.

AI in Education

- **Automated Content Generation:** With the help of Artificial intelligence, teachers can easily present quizzes, notes, and lesson plans so they do not waste time on manual typing, and the quality of the content is higher.
- **Virtual Tutors:** It is a feature that uses artificial intelligence and is available all the time to assist the learners with questions and guidelines.
- **Instant Feedback:** Since it can grade assignments and tests over the shortest time possible, it eases the process of checking on student's performance.
- **Personalized Learning Paths:** Through the help of an Artificial Intelligence application, students are provided with materials suiting their student's abilities and difficulties.

History of Artificial Intelligence (AI)

The word and technology are not new, and researchers are not entirely new to all the things artificial intelligence. You would think this technology is much older than you can imagine. In Ancient Greek and Egyptian myths, there are myths of Mechanical men. Here are some of the milestones of AI history that shaped the path of AI development between the present day and before:



Maturation of Artificial Intelligence (1943-1952)

There was plenty of progress in the development of artificial intelligence (AI) between 1943 and 1952. During that period, AI went from being a theory to a reality as experiments and practical applications were created. However, during this period, some of the key events were:

- **The Year 1943:** Warren McCulloch and Walter Pitts came up with the first work in 1943, which is now regarded as AI, and they presented a model of an artificial neuron.
- **The Year 1949:** Donald Hebb had shown an updating rule for the modification of the connection strength between neurons. However, this rule is now known as Hebbian learning.
- **The Year 1950:** "In 1950 Irishman Alan Turing published "Computing Machinery and Intelligence," where he suggested a test." The test might be used to test the machine's ability to display intelligent behaviour like that of human intelligence, a Turing test.
- **The Year 1951:**Initially, an artificial neural network (ANN), named SNARC, was created by Marvin Minsky and Dean Edmonds. For a network of 40 neurons, they used 3,000 vacuum tubes.

The birth of Artificial Intelligence (1952-1956)

Between 1952 and 1956, AI came to life as an interesting research field. It was during this time that pioneers and forward thinkers got together and laid the groundwork for what would eventually shape into an illimitable technological realm. Among this era's notable occurrences are the following:

- **Year 1952:** The world's first self-learning program for playing games was pioneered by Arthur Samuel of the Samuel Checkers Playing Program.
- **Year 1955:** "One of the first artificial intelligence programs, "According to the 41st IEEE Record of the 'Programmers First Artificial Intelligence Program' that was 'Logic Theorist,' Allen Newell and Herbert A. Simon created it.". Specifically, 38 of 52 Mathematics theorems have been proved, and new and more elegant proofs have been found for some theorems.
- **Year 1956:** First, the word Artificial Intelligence appeared in the context of the Dartmouth Conference (USA) by American Computer scientist John McCarthy. And then, for the first time, AI was coined as an academic field.

At this time, the invented high-level computer languages were such as FORTRAN, LISP, and COBOL. At that time, there was a great enthusiasm for AI.

The Golden Years- Early Enthusiasm (1956-1974)

The generally accepted period for artificial intelligence (AI) being in its Golden Age is from 1956 to 1974. This was an upswing of time for AI researchers and innovators; they had their enthusiasm and achieved tremendous advancements in this field. These are a few notable events of this era:

- **The Year 1958:** At this time, the perceptron, one of the first artificial neural networks that could learn from data, was introduced by Frank Rosenblatt. Modern neural networks were built on this invention. However, John McCarthy also developed the Lisp programming language, which quickly gained the favour of the AI community, and it turned into a very popular language among developers.
- **The Year 1959:** A seminal paper by Arthur Samuel, where he coined the term machine learning, and proposed that the computer would outperform their makers. Oliver Selfridge also had an important contribution to machine learning through his article titled 'Pandemonium: A Paradigm for Learning'. The results of this work are embodied in a model that can self-improve and discover patterns in events more quickly.
- **The Year 1964:** Daniel Bobrow, who is a doctoral candidate at MIT, saw it fit to create one of the early natural language processing (NLP) programs known as STUDENT, with the exact intent of solving algebra word problems.
- **The Year 1965:** Dendral was devised as the initial expert system by Edward Feigenbaum, Bruce G. Buchanan, Joshua Lederberg, and Carl Djerassi. The role of the test pattern in organic chemistry was to bias online organic chemists towards recognising unfamiliar organic compounds.
- **The Year 1966:** What the researchers aimed to develop were algorithms that could solve mathematical problems. ELIZA was the first chatbot that was created by Joseph Weizenbaum in 1966. Additionally, Stanford Research Institute developed the first

mobile intelligent robot that incorporated AI, computer vision, navigation, and NLP: Shakey. Today's self-driving cars and drones can be regarded as a precursor to this.

- **The Year 1968:** SHRDLU was developed by Terry Winograd, a pioneering multimodal AI that can understand the world of blocks and follow the user's instructions to manipulate and reason in that world.
- **The Year 1969:** A learning algorithm, backpropagation, was developed by Yu-Chi Ho and Arthur Bryson of which would lead to the ability to create multilayer artificial neural networks. It was a very important advance over the perceptron and the foundation for deep learning. Marvin Minsky and Seymour Papert also authored 'Perceptrons', which laid the framework for basic neural networks and the limitations of this. The results of this publication effectively put a halt to neural network research and lent credibility once again to the symbolic AI point of view.
- **The Year 1972:** WABOT-1 1 was the first intelligent humanoid robot built in Japan.
- **The Year 1973:** As a result, the British government drastically cut its backing of AI research on publication of the report Artificial Intelligence: A General Survey by James Lighthill.

The first AI winter (1974-1980)

A tough period for artificial intelligence (AI) is called the initial AI winter, which is between 1974 to 1980. At this time, there was a significant cut in research funding and a feeling of AI letdown.

- The first AI winter was from 1974 to 1980. AI winter is a term in computer science that means the period of severe shortage of funding from the government for AI research.
- It was during AI winters that the interest in publicity on artificial intelligence dropped.

A boom of AI (1980-1987)

After 1980, AI entered a golden period, its renaissance, its crown years, which, following the First AI winter, was a period of new vitality. These were some noteworthy events at this time:

- The first national conference of the American Association of Artificial Intelligence at Stanford University occurred in 1980.
- **The Year 1980:** AI came back after winter with 'Expert System.' The decision-making ability of a human expert was emulated using expert systems that were programmed. Meanwhile, AI also started another resurgence with Symbolics Lisp machines brought into use in the commercial world. However, in subsequent years, the Lisp machine market crashed.
- **The Year 1981:** Parallel computers for A.I. and other computational needs, based on architecture similar to that of current GPUs, were designed by Danny Hillis.
- **The Year 1984:** The phrase "AI winter" was first coined by Marvin Minsky and Roger Schank, as they spoke at an Association for the Advancement of Artificial Intelligence gathering. In turn, they cautioned the business world not to build up the expectations too high for AI, as this would end in disillusionment and the overall demise of the industry, which, alas, is exactly what happened three years later.
- **The Year 1985:** Bayesian network causal analysis was invented by Judea Pearl, who produced statistical methods for uncertainty in computer systems.

The second AI winter (1987-1993)

- The second AI Winter duration was from 1987 through to 1993.
- The high cost and inefficiency of the results again put the investors and the government to a stop, of funding AI research. The cost of using such an expert system as XCON was very cheap.

The emergence of intelligent agents (1993-2011)

From 1993 to 2011, artificial intelligence (AI) did quite a lot, especially with the development of intelligent computer programs. During this era, AI professionals' focus moved away from trying to imitate human intelligence to building pragmatic, ingenious software for specific tasks. Noteworthy occurrences at this period in time include:

- **The Year 1997:** To date, in 1997, IBM's Deep Blue won a historic victory against world chess champion Gary Kasparov, causing the first computer's victory over a reigning world chess champion. In addition, the Long Short-Term Memory recurrent neural network was introduced by Sepp Hochreiter and Jürgen Schmidhuber, allowing entire sequences of data, such as speech or video, to be processed.
- **The Year 2002:** The first home entry of AI was in Roomba, a vacuum cleaner.
- **The Year 2006:** Until 2006, companies like Facebook, Twitter, and Netflix also incorporated the use of AI into their respective business.
- **The Year 2009:** The paper Utilising Graphics Processors for Extensive Deep Unsupervised Learning was released by Rajat Raina, Anand Madhavan, and Andrew Ng to introduce the idea of GPUS being used to train large neural networks.
- **The Year 2011:** The initial CNN that achieved 'superhuman' performance came from Dan Claudiu Ciresan. Ueli Meier, Jürgen Schmidhuber, and Jonathan Masci solved the German Traffic Sign Recognition competition as the winners. Additionally, Apple released Siri, a voice-activated personal assistant that can process commands and generate responses that will then take place.

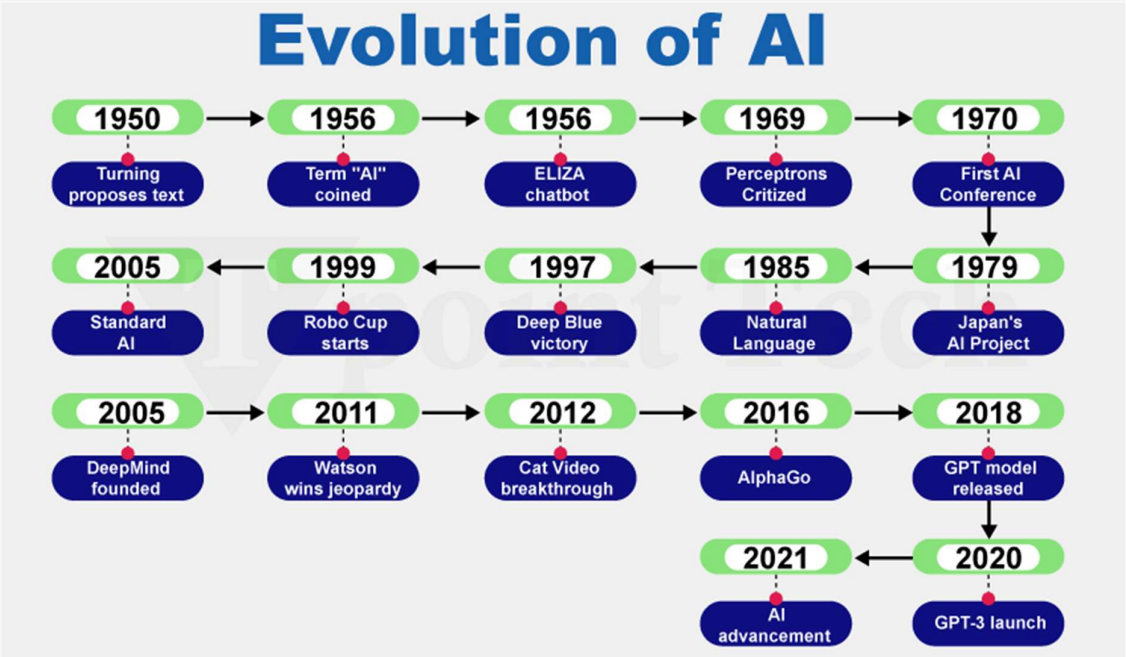
Deep Learning, Big Data, and Artificial General Intelligence (2011-present)

Significant progress has been witnessed in the artificial intelligence (AI) domain from 2011 to the present. They are the result of combining deep learning, putting data to use, and the never-ending hunt for artificial general intelligence (AGI). Some notable occurrences of this timeframe are therefore as follows:

- **The Year 2011:** In 2011, Watson was able to beat Jeopardy, a game show in which players were asked questions and riddles to answer. In other words, Watson was able to do things so quickly on the tricky questions, and it understood natural language so much already that it proved that it could.
- **The Year 2012:** To the user, Google has been launched with the 'Google Now' feature, which is similar to the way of Android app feature to predict information. Geoffrey Hinton, Ilya Sutskever, Alex Krizhevsky, etc., also introduced a deep CNN structure that won the ImageNet challenge, and their research resurgence brought deep learning back to the forefront.

- **The Year 2013:** Tianhe-2 system is the supercomputer in China that managed to double the world's fastest supercomputer's speed because it achieved a speed of 33.86 petaflops. The world's fastest system, it continued for the third year in a row. Second, DeepMind released other deep reinforcement learning models, a CNN trained on multiple trials using rewards, and succeeded in some games, beating humans. However, in addition to LeMons, Word2vec, the tool introduced by Microsoft researcher Tomas Mikolov and his team, does this automatically as well.
- **The Year 2014:** In the year 2014, Chatbot 'Eugene Goostman' won the infamous 'Turing test'. Diederik Kingma and Max Welling were on variational autoencoders (VAEs) that do images, images of videos, or text, while Ian Goodfellow and his group were the first to use generative adversarial networks (GANs) to make images, make images, wolf grinning masks, and deepfakes. Furthermore, Facebook's DeepFace deep learning based facial recognition system achieved similar (almost) human-level facial recognition (in images); they were able to recognise human faces in digital images with accuracy very similar to human beings.
- **The Year 2016:** This brings back the previous epoch when Kasparov played against Deep Blue, nearly 20 years ago, when AlphaGo defeated the Go world champion Lee Sedol in Seoul, South Korea. The company also pilots a self-driving car program for a limited group of users in Pittsburgh for one of its test grounds.
- **The Year 2018:** In fact, IBM's 'Project Debater' debated with two masters on complex topics and was good, very good indeed.
- Also known as Duplex, a Google AI program that booked phone hairdresser appointments tricked the lady on the other end of the line into thinking she was dealing with a machine.
- **The Year 2021:** The problem, however, is that Dall-E is an openAI multimodal AI system that can create images from the textual prompts.
- **The Year 2022:** However, last month, OpenAI launched GPT-3.5 LLM in a chat format - an alternative to presenting GPT-3.5 LLM.

Today, AI has come to great heights, but not so long ago. The concepts of Deep learning, Big data, and Data science have peaked a boom, and we can say the concepts are at par with a boom! The companies that have become partners with AI to develop marvelous devices today include Google, Facebook, IBM, and Amazon. So Artificial Intelligence's future is a high intelligence, which is an inspiring future.



Milestones in AI Development

One of the interesting things about the history of Artificial Intelligence (AI) is that it has been rich with achievements that helped the technology to evolve into one of the most transformative technologies that we are working on in the present.

Notable Achievements

AlphaGo

In 2016, a subsidiary of Google, Google's DeepMind, which develops AlphaGo, pulled off a big achievement for artificial intelligence by telling the world champion Go player Lee Sedol that it had beaten him.

- Given that a possible move in Go can be anything, and given the large number of possible moves, Go is a hard problem for computational systems to solve.
- For instance, AlphaGo had to synthesize deep learning and reinforcement learning to begin to understand strategies that had never been observed by humans before.
- AlphaGo's success in doing all this was the gateway to the development of other technologies in other fields (healthcare, logistics, etc.).

GPT Models (Generative Pre-trained Transformers)

NLP was revolutionised when machines were able to understand and create human-like text powered by OpenAI's GPT models.

- GPT-2 (2019), GPT-3 (2020), then the subsequent iterations have been used to establish benchmarks for language generation tasks.

- Since GPT models can be pre-trained on large datasets and then fine-tuned on specific tasks, they can be used in functions like chatbots, content creation, and assisting in coding processes.
- AI is used in industries of customer services, education, and entertainment, and this also has some ethical requirements for AI's AI-generated content.

DALL-E

DALL-E is also the second such generative AI for image synthesis that OpenAI developed.

- Different capability: It can generate a high-quality image from a text description by utilising image generation and language understanding.
- Between advertising, graphic design, and education, DALL-E is a swing of application areas and shows the growing homogenization of AI and the arts.
- Under the broader implication of multimodal AI systems, DALL-E has a wider implication, which combines a variety of data for sophisticated tasks.

International Collaboration in AI Research

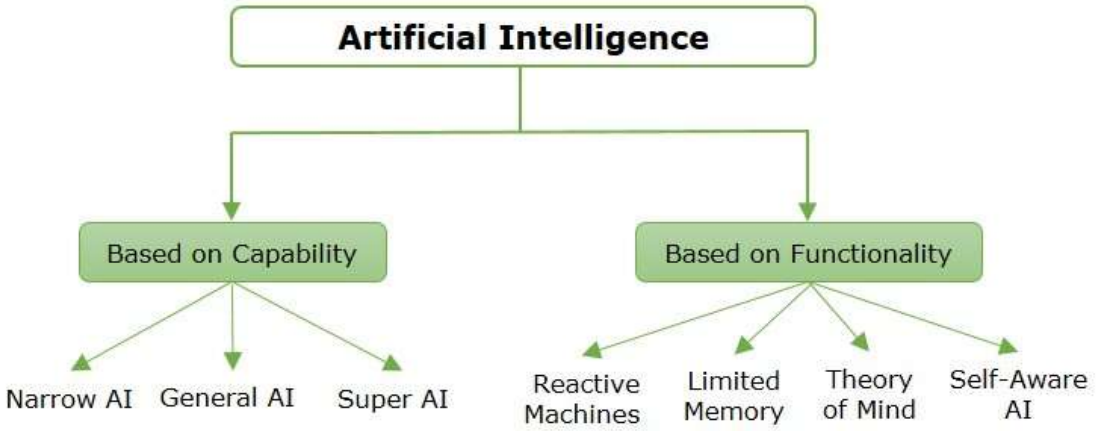
It is important for the fast development of AI on the part of researchers and organisations as well and governments due to global cooperation.

Collaborative Research Initiatives

- **Human Brain Project (EU)**: With models of cognition and learning inspired by neuroscience, it is intended to push the frontier of AIs.
- **Partnership on AI (Global)**: A collective of industry leaders and academia to first prevent the creation of AI without it being ethical and responsible.
- **AI for Good (UN)**: It wanted to employ AI in fighting against issues such as climate change, poverty, and healthcare.

Artificial Intelligence - Types

Artificial Intelligence (AI) is a technology that enables computers to think and act like humans. These systems are trained to learn from past experiences to enhance speed, precision, and effectiveness. Further based on the following criteria artificial intelligence can be categorized into different types −



Based on Capabilities

AI can be classified in different ways. One common classification is based on the **capabilities of the AI system**.

The three commonly discussed categories are:

1. **Narrow AI / Weak AI**
2. **Artificial General Intelligence (AGI)**
3. **Artificial Super Intelligence (ASI)**

The first category is widely used today. The other two are primarily concepts or goals of future AI research.

1. **Narrow AI / Weak AI**

Meaning

Narrow AI is AI designed to perform a **specific task or a limited set of related tasks**.

It can perform its designated task extremely well but does not possess general human-like intelligence.

Examples

- Face recognition
- Spam detection
- Product recommendation
- Voice assistants
- Fraud detection
- Navigation systems
- Chatbots
- Credit risk prediction

Business Example

An e-commerce website uses AI to recommend products.

For example:

A customer purchases a mobile phone → AI analyzes the customer's behavior → AI recommends a phone case and screen protector.

The system may be excellent at recommendations, but it cannot automatically become a business manager, doctor, or teacher without being specifically designed and trained for those tasks.

Characteristics

- Designed for specific tasks
- Limited scope
- Uses data and algorithms

- Can outperform humans in particular tasks
- Does not have general human intelligence
- Most AI applications available today fall into this category

Advantages

- High efficiency
- Fast processing
- Automation of repetitive tasks
- Can analyze large amounts of data
- Can reduce operational costs
- Supports business decisions

Limitation

A narrow AI system generally **does not transfer its intelligence automatically from one unrelated task to another**.

2. **Artificial General Intelligence (AGI)**

Meaning

Artificial General Intelligence (AGI) refers to a hypothetical AI system with broad intellectual abilities comparable to humans.

An AGI would theoretically be able to:

- Learn different types of tasks
- Reason across different situations
- Understand context
- Solve unfamiliar problems
- Transfer knowledge between tasks
- Adapt to new environments

Example

Imagine a machine that could:

1. Analyze a company's financial statements.
2. Develop a marketing strategy.
3. Teach a student accounting.
4. Learn a new language.
5. Plan a business trip.
6. Analyze a legal document.

—all while being able to learn and adapt broadly like a human.

That would be much closer to the idea of AGI than today's task-specific AI systems.

Important Point

AGI has not been established as a demonstrated reality. It remains a major research goal and a subject of debate regarding definitions and capabilities.

3. **Artificial Super Intelligence (ASI)**

Meaning

Artificial Super Intelligence (ASI) is a hypothetical form of AI that would possess intellectual abilities **far beyond those of humans across essentially all relevant domains**.

It is a theoretical concept rather than an existing technology.

Hypothetical Capabilities

ASI might theoretically be able to:

- Solve extremely complex scientific problems

- Learn much faster than humans
- Develop advanced technologies
- Analyze enormous amounts of information
- Make highly complex predictions
- Solve problems beyond human intellectual capabilities

Example

Imagine an AI system capable of outperforming the best human experts simultaneously in:

- Science
- Medicine
- Economics
- Engineering
- Mathematics
- Business strategy
- Creative problem-solving

Such a system would be closer to the concept of **ASI**.

Important Note

ASI is **hypothetical**. There is currently no established evidence that such a system exists.

Narrow AI vs AGI vs ASI

Feature	***Narrow AI***	***AGI***	***ASI***
Intelligence	Specialized	General	Beyond human level
Scope	Limited	Broad	Extremely broad
Current status	Widely used	Not established	Hypothetical
Learning	Task-specific	General-purpose	Potentially highly advanced

Example	Recommendation system	Human-like general intelligence	Theoretical superintelligence
Business use	Common	Future possibility	Hypothetical
Human-level ability	Only in specific tasks	Broadly human-like	Exceeds humans

Based on Functionality

AI is classified into the following types based on the functionality −

Reactive Machines

Reactive Machines are the most basic type of artificial intelligence. These machines operate only on the present data and do not store any previous experiences or learn from past actions. Additionally, these systems respond to specific inputs with predetermined outputs and cannot be changed.

IBM's Deep Blue is a great example of reactive machines. It is the first computer system to defeat a reigning world chess champion, Garry Kasparov. It could identify pieces on the board and make predictions but could not store any memories or learn from the past games.

Google's AlphaGo is another example of a reactive machine, playing the board game Go with a similar method of pattern recognition without gaining knowledge from previous games.

Limited Memory

Limited Memory is the most used category in most modern AI applications. It can store past experiences and learn from them to improve future outcomes. These machines store historical data to predict and make decisions but do not have long-term memory. Major applications like **autonomous systems** and **robotics** often rely on limited memory.

Chatbots is an example of limited memory, where it can remember recent conversations to improve the flow and relevance. Additionally, **self-driving cars** is another example that observes the road, traffic signs, and surroundings to make decisions based on past experiences and current conditions.

Theory of Mind

Theory of Mind could understand the human emotions, beliefs, and intentions. While this type of AI is still in development, it has enabled machines to interpret emotions accurately and modify behavior accordingly so that machines could interact with humans effectively. Some of the possible applications of this type are probably **collaborating robots** and **human-robot interaction.**

Self-Awareness

Self-Aware AI represents the future of artificial intelligence with self-consciousness and awareness similar to humans.While we are far from achieving the goal of self-aware AI, it is an important objective for the development of AI. The applications of self-aware AI could be fully autonomous systems that could take moral and ethical decisions.

Artificial intelligence vs Machine Learning vs Deep Learning

We often confuse the terms Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL), assuming they all mean the same thing. But in reality, they represent different levels of intelligent computing. AI is the broadest field, ML is a subset of AI and DL is a further subset of ML.

1. Artificial Intelligence (AI)

Artificial intelligence is the field of computer science focused on creating systems that can perform tasks requiring human-like intelligence, such as reasoning, problem-solving and language understanding. AI can use rule-based methods as well as machine learning.

- **Examples**: Sophia robot, AI chatbots, fraud detection, medical diagnosis.

2. Machine Learning (ML)

Machine Learning is a branch of AI that enables systems to learn patterns from data and use them to make predictions or decisions. Common approaches include supervised, unsupervised and reinforcement learning.

- **Examples**: Amazon recommendations, Netflix suggestions, spam detection.

3. Deep Learning (DL)

Deep Learning is a subset of machine learning that uses multi-layer neural networks to learn complex patterns from data. It is widely used for tasks involving images, audio, text and video.

- **Examples**: Self-driving cars, facial recognition, voice assistants.

AI vs. ML vs. DL

Aspect	AI	ML	DL
Definition	Broad field of creating intelligent systems	Subfield of AI that learns from data	Subfield of ML based on deep neural networks
Scope	Broadest	Narrower than AI	Narrowest

Aspect	AI	ML	DL
How It Works	Uses rules, search, ML, DL and other techniques	Learns patterns from data using algorithms	Uses multi-layer neural networks to learn complex patterns
Data Requirements	Depends on the approach	Generally requires training data	Often benefits from large datasets, especially for complex tasks
Feature Engineering	Depends on the approach	Often requires manually selected or engineered features	Can automatically learn features from raw inputs
Common Techniques	Expert systems, search, ML, DL and reinforcement learning	Regression, decision trees, clustering, SVMs and ensemble methods	CNNs, RNNs, LSTMs, Transformers and other neural network architectures
Applications	Robotics, planning, NLP, expert systems and intelligent assistants	Classification, regression, recommendation and forecasting	Computer vision, speech recognition and generative AI
Computational Requirements	Varies widely	Usually moderate, depending on the algorithm and dataset	Often high, particularly when training large models

Role of AI in Data Analytics

Artificial Intelligence (AI) has revolutionized the field of data analytics, providing powerful tools and techniques to extract valuable insights from vast amounts of data. By leveraging AI, organizations can enhance their decision-making processes, optimize operations, and gain a competitive edge in the market. In this article, we will explore the concepts of AI in data analytics, its applications, benefits, challenges, and future trends.

What is Artificial Intelligence in Data Analytics?
Artificial Intelligence refers to the simulation of human intelligence in machines that are programmed to think and act like humans. In the context of data analytics, AI algorithms analyze large datasets, identify patterns, and make decisions with minimal human intervention. AI in data analytics encompasses various subfields, including machine learning, deep learning, natural language processing (NLP), and computer vision.

Applications of AI in Data Analytics

1. **Predictive Analytics**: AI algorithms analyze historical data to predict future trends and outcomes. For example, AI can forecast sales, customer behavior, and equipment maintenance needs.
2. **Prescriptive Analytics**: AI recommends actions to optimize outcomes based on predictive analytics. For instance, AI can suggest pricing strategies or inventory management tactics.
3. **Anomaly Detection**: AI detects outliers or anomalies in data that deviate from normal patterns, indicating potential issues or opportunities.
4. **Natural Language Processing (NLP)**: AI enables computers to understand, interpret, and generate human language. NLP is used in sentiment analysis, chatbots, and text summarization.
5. **Computer Vision**: AI algorithms analyze and interpret visual information from images or videos. Applications include facial recognition, object detection, and autonomous vehicles.
6. **Fraud Detection**: AI helps identify fraudulent activities by analyzing patterns and anomalies in financial transactions or user behavior.

Benefits of AI in Data Analytics

1. **Improved Decision Making**: AI provides accurate and timely insights, enabling organizations to make informed decisions.
2. **Cost Efficiency**: AI automates repetitive tasks, reducing operational costs and increasing efficiency.
3. **Enhanced Customer Experience**: AI-powered analytics can personalize customer interactions, leading to higher satisfaction and loyalty.
4. **Increased Productivity**: AI algorithms can process data faster than humans, leading to faster insights and decision-making.

5. **Risk Management**: AI helps identify and mitigate risks by analyzing data patterns and anomalies.

Challenges of AI in Data Analytics

1. **Data Quality**: AI models require high-quality data for accurate predictions. Poor data quality can lead to biased or unreliable results.
2. **Data Privacy**: AI raises concerns about data privacy and security, as it often requires access to sensitive information.
3. **Interpretability**: AI models can be complex and difficult to interpret, making it challenging to understand the reasoning behind their decisions.
4. **Ethical Concerns**: AI algorithms can reinforce biases present in the data, leading to discriminatory outcomes.
5. **Implementation Complexity**: Integrating AI into existing data analytics workflows can be complex and require specialized skills.

Future Trends in AI for Data Analytics

1. **Automated Machine Learning (AutoML)**: AutoML tools simplify the process of building and deploying AI models, making AI more accessible to non-experts.
2. **Explainable AI**: Efforts are underway to make AI algorithms more transparent and explainable, addressing concerns about bias and ethics.
3. **Edge Computing**: AI is being deployed on edge devices to process data locally, reducing latency and enhancing privacy.
4. **AI Ethics and Governance**: There is a growing focus on establishing ethical guidelines and governance frameworks for AI to ensure responsible use.
5. **Augmented Analytics**: Augmented analytics combines AI and human intelligence to enhance data analysis capabilities.

In conclusion, AI is transforming the field of data analytics, enabling organizations to unlock the full potential of their data. By leveraging AI technologies, businesses can gain valuable insights, improve decision-making, and stay competitive in today's data-driven world.